Eureka Math（A Story of Ratios）6–8 大纲与海报计划

版本：Eureka Math™ A Story of Ratios（Great Minds 编写，即纽约州 EngageNY 免费发布的 6–8 年级数学模块），依据 Great Minds 官网免费提供的 Eureka Basic Curriculum Files 中的各模块 Teacher Edition PDF（版本号 1.3.x，2015–2016 年印次，© 2015 Great Minds，非商业使用依 CC BY-NC-SA 4.0 授权）。模块、主题（Topic）、课时的英文标题和每个 Topic 对应的课程标准逐字摘自各 Teacher Edition 的目录，没有自行编写。

海报粒度：和 K–5 一样每个 Topic 一张。全套 71 个 Topic（六年级 27、七年级 23、八年级 21），所以计划 71 张海报。

本地化：英制单位改为公制，美元改为人民币，涉及的 Topic 下面有标注。海报里自己举的例子都标了「示例」；动手活动加了安全提示。

来源

- Great Minds：A Story of Ratios: Eureka Basic Curriculum Files（各模块 Teacher Edition PDF，下面每个模块都给了链接）
- 课程总览：A Story of Ratios Curriculum Overview（PDF） · Curriculum Map（PDF）

六年级（Grade 6）· 6 个模块 · 27 个 Topic

第1模块 比和单位比率（Module 1: Ratios and Unit Rates）

- 来源：Teacher Edition PDF

A主题 比的表示和推理（Topic A: Representing and Reasoning About Ratios）

- 课程标准：6.RP.A.1, 6.RP.A.3a
- Lesson 1–2：Ratios
- Lesson 3–4：Equivalent Ratios
- Lesson 5–6：Solving Problems by Finding Equivalent Ratios
- Lesson 7：Associated Ratios and the Value of a Ratio
- Lesson 8：Equivalent Ratios Defined Through the Value of a Ratio
- 海报要点：比是一对有序的数；等值比；比值

B主题 等值比的集合（Topic B: Collections of Equivalent Ratios）

- 课程标准：6.RP.A.3a
- Lesson 9：Tables of Equivalent Ratios
- Lesson 10：The Structure of Ratio Tables—Additive and Multiplicative
- Lesson 11：Comparing Ratios Using Ratio Tables
- Lesson 12：From Ratio Tables to Double Number Line Diagrams
- Lesson 13：From Ratio Tables to Equations Using the Value of a Ratio
- Lesson 14：From Ratio Tables, Equations, and Double Number Line Diagrams to Plots on the Coordinate Plane
- Lesson 15：A Synthesis of Representations of Equivalent Ratio Collections
- 海报要点：比率表、双数线、方程、坐标图四种表示

C主题 单位比率（Topic C: Unit Rates）

- 课程标准：6.RP.A.2, 6.RP.A.3b, 6.RP.A.3d
- Lesson 16：From Ratios to Rates
- Lesson 17：From Rates to Ratios
- Lesson 18：Finding a Rate by Dividing Two Quantities
- Lesson 19–20：Comparison Shopping—Unit Price and Related Measurement Conversions
- Lesson 21–22：Getting the Job Done—Speed, Work, and Measurement Units
- Lesson 23：Problem-Solving Using Rates, Unit Rates, and Conversions
- 海报要点：从比到比率；单价比较；速度、工作效率；单位换算
- 本地化/安全：英制单位换算改为公制（千米、米、厘米、千克、升），价格用人民币

D主题 百分数（Topic D: Percent）

- 课程标准：6.RP.A.3c
- Lesson 24：Percent and Rates per 100
- Lesson 25：A Fraction as a Percent
- Lesson 26：Percent of a Quantity
- Lesson 27–29：Solving Percent Problems

- 海报要点：百分数是每 100 中的几；分数化百分数；求部分、整体、百分比
- 本地化/安全：价格用人民币

第2模块 分数除法与小数运算（Module 2: Arithmetic Operations Including Division of Fractions）

- 来源：Teacher Edition PDF

A主题 分数除以分数（Topic A: Dividing Fractions by Fractions）

- 课程标准：6.NS.A.1
- Lesson 1：Interpreting Division of a Fraction by a Whole Number—Visual Models
- Lesson 2：Interpreting Division of a Whole Number by a Fraction—Visual Models
- Lesson 3–4：Interpreting and Computing Division of a Fraction by a Fraction—More Models
- Lesson 5：Creating Division Stories
- Lesson 6：More Division Stories
- Lesson 7：The Relationship Between Visual Fraction Models and Equations
- Lesson 8：Dividing Fractions and Mixed Numbers
- 海报要点：图示分数除法；编除法故事；除以一个分数等于乘它的倒数

B主题 小数加减乘（Topic B: Multi-Digit Decimal Operations—Adding, Subtracting, and Multiplying）

- 课程标准：6.NS.B.3
- Lesson 9：Sums and Differences of Decimals
- Lesson 10：The Distributive Property and the Products of Decimals
- Lesson 11：Fraction Multiplication and the Products of Decimals
- 海报要点：数位对齐加减；分配律与小数乘法；用分数理解积的小数位

C主题 整数和小数的除法（Topic C: Dividing Whole Numbers and Decimals）

- 课程标准：6.NS.B.2, 6.NS.B.3
- Lesson 12：Estimating Digits in a Quotient
- Lesson 13：Dividing Multi-Digit Numbers Using the Algorithm
- Lesson 14：The Division Algorithm—Converting Decimal Division into Whole Number Division Using Fractions
- Lesson 15：The Division Algorithm—Converting Decimal Division into Whole Number Division Using Mental Math
- 海报要点：估商；长除法竖式；把小数除法化成整数除法

D主题 数论（Topic D: Number Theory—Thinking Logically About Multiplicative Arithmetic）

- 课程标准：6.NS.B.4
- Lesson 16：Even and Odd Numbers
- Lesson 17：Divisibility Tests for 3 and 9
- Lesson 18：Least Common Multiple and Greatest Common Factor
- Lesson 19：The Euclidean Algorithm as an Application of the Long Division Algorithm
- 海报要点：奇偶数；被 3、9 整除；最小公倍数和最大公因数；辗转相除法

第3模块 有理数（Module 3: Rational Numbers）

- 来源：Teacher Edition PDF

A主题 数轴上的正数和负数（Topic A: Understanding Positive and Negative Numbers on the Number Line）

- 课程标准：6.NS.C.5, 6.NS.C.6a, 6.NS.C.6c
- Lesson 1：Positive and Negative Numbers on the Number Line—Opposite Direction and Value
- Lesson 2–3：Real-World Positive and Negative Numbers and Zero
- Lesson 4：The Opposite of a Number
- Lesson 5：The Opposite of a Number’ s Opposite
- Lesson 6：Rational Numbers on the Number Line
- 海报要点：相反方向和数值；生活中的正负数；相反数；有理数在数轴上

B主题 顺序和绝对值（Topic B: Order and Absolute Value）

- 课程标准：6.NS.C.6c, 6.NS.C.7
- Lesson 7–8：Ordering Integers and Other Rational Numbers
- Lesson 9：Comparing Integers and Other Rational Numbers
- Lesson 10：Writing and Interpreting Inequality Statements Involving Rational Numbers
- Lesson 11：Absolute Value—Magnitude and Distance
- Lesson 12：The Relationship Between Absolute Value and Order
- Lesson 13：Statements of Order in the Real World
- 海报要点：比较和排序；不等式；绝对值是到 0 的距离

C主题 有理数与坐标平面（Topic C: Rational Numbers and the Coordinate Plane）

- 课程标准：6.NS.C.6b, 6.NS.C.6c, 6.NS.C.8
- Lesson 14：Ordered Pairs
- Lesson 15：Locating Ordered Pairs on the Coordinate Plane
- Lesson 16：Symmetry in the Coordinate Plane
- Lesson 17：Drawing the Coordinate Plane and Points on the Plane
- Lesson 18：Distance on the Coordinate Plane
- Lesson 19：Problem Solving and the Coordinate Plane
- 海报要点：有序数对；四个象限；对称点；坐标平面上的距离

第4模块 代数式和方程（Module 4: Expressions and Equations）

- 来源：Teacher Edition PDF

A主题 运算之间的关系（Topic A: Relationships of the Operations）

- 课程标准：6.EE.A.3
- Lesson 1：The Relationship of Addition and Subtraction
- Lesson 2：The Relationship of Multiplication and Division
- Lesson 3：The Relationship of Multiplication and Addition
- Lesson 4：The Relationship of Division and Subtraction
- 海报要点：加减互逆、乘除互逆；乘法是重复加；除法是重复减

B主题 运算的特殊记法（Topic B: Special Notations of Operations）

- 课程标准：6.EE.A.1, 6.EE.A.2c
- Lesson 5：Exponents
- Lesson 6：The Order of Operations

- 海报要点：指数表示重复乘；运算顺序

C主题 字母与数（Topic C: Replacing Letters and Numbers）

- 课程标准：6.EE.A.2c, 6.EE.A.4
- Lesson 7：Replacing Letters with Numbers
- Lesson 8：Replacing Numbers with Letters
- 海报要点：用数代替字母，用字母代替数

D主题 展开、因式分解与分配（Topic D: Expanding, Factoring, and Distributing Expressions）

- 课程标准：6.EE.A.2a, 6.EE.A.2b, 6.EE.A.3, 6.EE.A.4
- Lesson 9：Writing Addition and Subtraction Expressions
- Lesson 10：Writing and Expanding Multiplication Expressions
- Lesson 11：Factoring Expressions
- Lesson 12：Distributing Expressions
- Lesson 13–14：Writing Division Expressions
- 海报要点：写加减乘除式；分配律展开和提取公因数

E主题 用代数式表示运算（Topic E: Expressing Operations in Algebraic Form）

- 课程标准：6.EE.A.2a, 6.EE.A.2b
- Lesson 15：Read Expressions in Which Letters Stand for Numbers
- Lesson 16–17：Write Expressions in Which Letters Stand for Numbers
- 海报要点：读、写含字母的式子：和、差、积、商

F主题 写式子和求值（Topic F: Writing and Evaluating Expressions and Formulas）

- 课程标准：6.EE.A.2a, 6.EE.A.2c, 6.EE.B.6
- Lesson 18：Writing and Evaluating Expressions—Addition and Subtraction
- Lesson 19：Substituting to Evaluate Addition and Subtraction Expressions
- Lesson 20：Writing and Evaluating Expressions—Multiplication and Division
- Lesson 21：Writing and Evaluating Expressions—Multiplication and Addition
- Lesson 22：Writing and Evaluating Expressions—Exponents
- 海报要点：列式并代入求值；公式

G主题 解方程（Topic G: Solving Equations）

- 课程标准：6.EE.B.5, 6.EE.B.6, 6.EE.B.7
- Lesson 23–24：True and False Number Sentences
- Lesson 25：Finding Solutions to Make Equations True
- Lesson 26：One-Step Equations—Addition and Subtraction
- Lesson 27：One-Step Equations—Multiplication and Division
- Lesson 28：Two-Step Problems—All Operations
- Lesson 29：Multi-Step Problems—All Operations
- 海报要点：真假数句；使等式成立的解；一步和两步方程

H主题 方程的应用（Topic H: Applications of Equations）

- 课程标准：6.EE.B.5, 6.EE.B.6, 6.EE.B.7, 6.EE.B.8, 6.EE.C.9
- Lesson 30：One-Step Problems in the Real World
- Lesson 31：Problems in Mathematical Terms

- Lesson 32：Multi-Step Problems in the Real World
- Lesson 33：From Equations to Inequalities
- Lesson 34：Writing and Graphing Inequalities in Real-World Problems
- 海报要点：列方程解应用题；从方程到不等式，画在数轴上

第5模块 面积、表面积和体积（Module 5: Area, Surface Area, and Volume Problems）

- 来源：Teacher Edition PDF

A主题 三角形和多边形的面积（Topic A: Area of Triangles, Quadrilaterals, and Polygons）

- 课程标准：6.G.A.1
- Lesson 1：The Area of Parallelograms Through Rectangle Facts
- Lesson 2：The Area of Right Triangles
- Lesson 3：The Area of Acute Triangles Using Height and Base
- Lesson 4：The Area of All Triangles Using Height and Base
- Lesson 5：The Area of Polygons Through Composition and Decomposition
- Lesson 6：Area in the Real World
- 海报要点：平行四边形、三角形面积；多边形分割与组合

B主题 坐标平面上的多边形（Topic B: Polygons on the Coordinate Plane）

- 课程标准：6.G.A.3
- Lesson 7：Distance on the Coordinate Plane
- Lesson 8：Drawing Polygons in the Coordinate Plane
- Lesson 9：Determining Perimeter and Area of Polygons on the Coordinate Plane
- Lesson 10：Distance, Perimeter, and Area in the Real World
- 海报要点：坐标距离；画多边形；求周长和面积

C主题 长方体的体积（Topic C: Volume of Right Rectangular Prisms）

- 课程标准：6.G.A.2
- Lesson 11：Volume with Fractional Edge Lengths and Unit Cubes
- Lesson 12：From Unit Cubes to the Formulas for Volume
- Lesson 13：The Formulas for Volume
- Lesson 14：Volume in the Real World
- 海报要点：分数棱长的单位立方体；V = 长 × 宽 × 高 = 底面积 × 高
- 本地化/安全：英寸、英尺改为厘米、米

D主题 展开图和表面积（Topic D: Nets and Surface Area）

- 课程标准：6.G.A.2, 6.G.A.4
- Lesson 15：Representing Three-Dimensional Figures Using Nets
- Lesson 16：Constructing Nets
- Lesson 17：From Nets to Surface Area
- Lesson 18：Determining Surface Area of Three-Dimensional Figures
- Lesson 19：Surface Area and Volume in the Real World
- 海报要点：立体图形的展开图；由展开图求表面积

第6模块 统计（Module 6: Statistics）

- 来源：Teacher Edition PDF

A主题 理解分布（Topic A: Understanding Distributions）

- 课程标准：6.SP.A.1, 6.SP.A.2, 6.SP.B.4, 6.SP.B.5b
- Lesson 1：Posing Statistical Questions
- Lesson 2：Displaying a Data Distribution
- Lesson 3：Creating a Dot Plot
- Lesson 4：Creating a Histogram
- Lesson 5：Describing a Distribution Displayed in a Histogram
- 海报要点：统计问题；点图；频数表和直方图；描述分布形状

B主题 平均数和平均绝对偏差（Topic B: Summarizing a Distribution That Is Approximately Symmetric Using the Mean and Mean Absolute Deviation）

- 课程标准：6.SP.A.2, 6.SP.A.3, 6.SP.B.4, 6.SP.B.5
- Lesson 6：Describing the Center of a Distribution Using the Mean
- Lesson 7：The Mean as a Balance Point
- Lesson 8：Variability in a Data Distribution
- Lesson 9：The Mean Absolute Deviation (MAD)
- Lesson 10–11：Describing Distributions Using the Mean and MAD
- 海报要点：平均数是平衡点；变异性；平均绝对偏差（MAD）

C主题 中位数和四分位距（Topic C: Summarizing a Distribution That Is Skewed Using the Median and the Interquartile Range）

- 课程标准：6.SP.A.2, 6.SP.A.3, 6.SP.B.4, 6.SP.B.5
- Lesson 12：Describing the Center of a Distribution Using the Median
- Lesson 13：Describing Variability Using the Interquartile Range (IQR)
- Lesson 14：Summarizing a Distribution Using a Box Plot
- Lesson 15：More Practice with Box Plots
- Lesson 16：Understanding Box Plots
- 海报要点：中位数；四分位距（IQR）；箱线图

D主题 概括和描述分布（Topic D: Summarizing and Describing Distributions）

- 课程标准：6.SP.B.4, 6.SP.B.5
- Lesson 17：Developing a Statistical Project
- Lesson 18：Connecting Graphical Representations and Numerical Summaries
- Lesson 19：Comparing Data Distributions
- Lesson 20：Describing Center, Variability, and Shape of a Data Distribution from a Graphical Representation
- Lesson 21：Summarizing a Data Distribution by Describing Center, Variability, and Shape
- Lesson 22：Presenting a Summary of a Statistical Project
- 海报要点：统计项目；比较分布；描述中心、变异性和形状
- 本地化/安全：统计项目只用班级自愿提供的、不涉及隐私的数据

七年级（Grade 7）· 6 个模块 · 23 个 Topic

第1模块 比例关系（Module 1: Ratios and Proportional Relationships）

- 来源：Teacher Edition PDF

A主题 比例关系（Topic A: Proportional Relationships）

- 课程标准：7.RP.A.2a
- Lesson 1：An Experience in Relationships as Measuring Rate
- Lesson 2：Proportional Relationships
- Lesson 3–4：Identifying Proportional and Non-Proportional Relationships in Tables
- Lesson 5–6：Identifying Proportional and Non-Proportional Relationships in Graphs
- 海报要点：用表格判断是不是比例关系

B主题 单位比率和比例常数（Topic B: Unit Rate and the Constant of Proportionality）

- 课程标准：7.RP.A.2b, 7.RP.A.2c, 7.RP.A.2d, 7.EE.B.4a
- Lesson 7：Unit Rate as the Constant of Proportionality
- Lesson 8–9：Representing Proportional Relationships with Equations
- Lesson 10：Interpreting Graphs of Proportional Relationships
- 海报要点：单位比率就是比例常数 k；y = kx；图像过原点

C主题 含分数的比和比率（Topic C: Ratios and Rates Involving Fractions）

- 课程标准：7.RP.A.1, 7.RP.A.3, 7.EE.B.4a
- Lesson 11–12：Ratios of Fractions and Their Unit Rates
- Lesson 13：Finding Equivalent Ratios Given the Total Quantity
- Lesson 14：Multi-Step Ratio Problems
- Lesson 15：Equations of Graphs of Proportional Relationships Involving Fractions
- 海报要点：分数的比与单位比率；已知总量求各部分；多步问题

D主题 比例尺图（Topic D: Ratios of Scale Drawings）

- 课程标准：7.RP.A.2b, 7.G.A.1
- Lesson 16：Relating Scale Drawings to Ratios and Rates
- Lesson 17：The Unit Rate as the Scale Factor
- Lesson 18：Computing Actual Lengths from a Scale Drawing
- Lesson 19：Computing Actual Areas from a Scale Drawing
- Lesson 20：An Exercise in Creating a Scale Drawing
- Lesson 21–22：An Exercise in Changing Scales
- 海报要点：比例尺；由图上长度求实际长度和面积；改变比例尺
- 本地化/安全：比例尺图用厘米和米

第2模块 有理数（Module 2: Rational Numbers）

- 来源：Teacher Edition PDF

A主题 有理数的加减（Topic A: Addition and Subtraction of Integers and Rational Numbers）

- 课程标准：7.NS.A.1

- Lesson 1：Opposite Quantities Combine to Make Zero
- Lesson 2：Using the Number Line to Model the Addition of Integers
- Lesson 3：Understanding Addition of Integers
- Lesson 4：Efficiently Adding Integers and Other Rational Numbers
- Lesson 5：Understanding Subtraction of Integers and Other Rational Numbers
- Lesson 6：The Distance Between Two Rational Numbers
- Lesson 7：Addition and Subtraction of Rational Numbers
- Lesson 8–9：Applying the Properties of Operations to Add and Subtract Rational Numbers
- 海报要点：相反数相加得 0；数轴上的加法；减去一个数等于加它的相反数；距离

B主题 有理数的乘除（Topic B: Multiplication and Division of Integers and Rational Numbers）

- 课程标准：7.NS.A.2
- Lesson 10：Understanding Multiplication of Integers
- Lesson 11：Develop Rules for Multiplying Signed Numbers
- Lesson 12：Division of Integers
- Lesson 13：Converting Between Fractions and Decimals Using Equivalent Fractions
- Lesson 14：Converting Rational Numbers to Decimals Using Long Division
- Lesson 15：Multiplication and Division of Rational Numbers
- Lesson 16：Applying the Properties of Operations to Multiply and Divide Rational Numbers
- 海报要点：整数乘法的符号规则；整数除法；分数和小数互化

C主题 有理数运算的应用（Topic C: Applying Operations with Rational Numbers to Expressions and Equations）

- 课程标准：7.NS.A.3, 7.EE.A.2, 7.EE.B.4a
- Lesson 17：Comparing Tape Diagram Solutions to Algebraic Solutions
- Lesson 18–19：Writing, Evaluating, and Finding Equivalent Expressions with Rational Numbers
- Lesson 20：Investments—Performing Operations with Rational Numbers
- Lesson 21：If-Then Moves with Integer Number Cards
- Lesson 22–23：Solving Equations Using Algebra
- 海报要点：线段图与代数解法；有理数的式子和方程

第3模块 代数式和方程（Module 3: Expressions and Equations）

- 来源：Teacher Edition PDF

A主题 等价代数式（Topic A: Use Properties of Operations to Generate Equivalent Expressions）

- 课程标准：7.EE.A.1, 7.EE.A.2
- Lesson 1–2：Generating Equivalent Expressions
- Lesson 3–4：Writing Products as Sums and Sums as Products
- Lesson 5：Using the Identity and Inverse to Write Equivalent Expressions
- Lesson 6：Collecting Rational Number Like Terms
- 海报要点：用运算律写等价式；合并同类项；乘积与和互化

B主题 方程与不等式（Topic B: Solve Problems Using Expressions, Equations, and Inequalities）

- 课程标准：7.EE.B.3, 7.EE.B.4, 7.G.B.5
- Lesson 7：Understanding Equations

- Lesson 8–9：Using If-Then Moves in Solving Equations
- Lesson 10–11：Angle Problems and Solving Equations
- Lesson 12：Properties of Inequalities
- Lesson 13：Inequalities
- Lesson 14：Solving Inequalities
- Lesson 15：Graphing Solutions to Inequalities
- 海报要点：等式性质（「如果…那么…」）；角的方程；解不等式并画图

C主题 几何问题（Topic C: Use Equations and Inequalities to Solve Geometry Problems）

- 课程标准：7.G.B.4, 7.G.B.6
- Lesson 16：The Most Famous Ratio of All
- Lesson 17：The Area of a Circle
- Lesson 18：More Problems on Area and Circumference
- Lesson 19：Unknown Area Problems on the Coordinate Plane
- Lesson 20：Composite Area Problems
- Lesson 21–22：Surface Area
- Lesson 23–24：The Volume of a Right Prism
- Lesson 25–26：Volume and Surface Area
- 海报要点：圆周率；圆的面积；组合图形面积；表面积；直棱柱体积

第4模块 百分数和比例关系（Module 4: Percent and Proportional Relationships）

- 来源：Teacher Edition PDF

A主题 求整体（Topic A: Finding the Whole）

- 课程标准：7.RP.A.1, 7.RP.A.2c, 7.RP.A.3
- Lesson 1：Percent
- Lesson 2：Part of a Whole as a Percent
- Lesson 3：Comparing Quantities with Percent
- Lesson 4：Percent Increase and Decrease
- Lesson 5：Finding One Hundred Percent Given Another Percent
- Lesson 6：Fluency with Percents
- 海报要点：百分数；百分比比较；增加和减少；已知部分求整体

B主题 多个整体的百分数问题（Topic B: Percent Problems Including More Than One Whole）

- 课程标准：7.RP.A.1, 7.RP.A.2, 7.RP.A.3, 7.EE.B.3
- Lesson 7：Markup and Markdown Problems
- Lesson 8：Percent Error Problems
- Lesson 9：Problem Solving When the Percent Changes
- Lesson 10：Simple Interest
- Lesson 11：Tax, Commissions, Fees, and Other Real-World Percent Applications
- 海报要点：加价和打折；百分误差；单利；税费等应用
- 本地化/安全：美元改为人民币；美国销售税改为一般的「税费、服务费」说法

C主题 比例尺图（Topic C: Scale Drawings）

- 课程标准：7.RP.A.2b, 7.G.A.1

- Lesson 12：The Scale Factor as a Percent for a Scale Drawing
- Lesson 13：Changing Scales
- Lesson 14：Computing Actual Lengths from a Scale Drawing
- Lesson 15：Solving Area Problems Using Scale Drawings
- 海报要点：比例尺写成百分数；改变比例尺；求实际长度和面积

D主题 人口、混合和计数问题（Topic D: Population, Mixture, and Counting Problems Involving Percents）

- 课程标准：7.RP.A.2c, 7.RP.A.3, 7.EE.B.3
- Lesson 16：Population Problems
- Lesson 17：Mixture Problems
- Lesson 18：Counting Problems
- 海报要点：人口百分比；溶液浓度；计数问题
- 本地化/安全：混合问题加安全提示：溶液实验只用清水和食用色素，不尝

第5模块 统计与概率（Module 5: Statistics and Probability）

- 来源：Teacher Edition PDF

A主题 计算和理解概率（Topic A: Calculating and Interpreting Probabilities）

- 课程标准：7.SP.C.5, 7.SP.C.6, 7.SP.C.7, 7.SP.C.8a, 7.SP.C.8b
- Lesson 1：Chance Experiments
- Lesson 2：Estimating Probabilities by Collecting Data
- Lesson 3：Chance Experiments with Equally Likely Outcomes
- Lesson 4：Calculating Probabilities for Chance Experiments with Equally Likely Outcomes
- Lesson 5：Chance Experiments with Outcomes That Are Not Equally Likely
- Lesson 6：Using Tree Diagrams to Represent a Sample Space and to Calculate Probabilities
- Lesson 7：Calculating Probabilities of Compound Events
- 海报要点：概率实验；等可能结果；树状图；复合事件
- 本地化/安全：实验用硬币、骰子、转盘

B主题 估计概率（Topic B: Estimating Probabilities）

- 课程标准：7.SP.C.6, 7.SP.C.7, 7.SP.C.8c
- Lesson 8：The Difference Between Theoretical Probabilities and Estimated Probabilities
- Lesson 9：Comparing Estimated Probabilities to Probabilities Predicted by a Model
- Lesson 10–11：Conducting a Simulation to Estimate the Probability of an Event
- Lesson 12：Applying Probability to Make Informed Decisions
- 海报要点：理论概率和估计概率；模拟实验；用概率做决定

C主题 随机抽样与估计总体（Topic C: Random Sampling and Estimating Population Characteristics）

- 课程标准：7.SP.A.1, 7.SP.A.2
- Lesson 13：Populations, Samples, and Generalizing from a Sample to a Population
- Lesson 14：Selecting a Sample
- Lesson 15：Random Sampling
- Lesson 16：Methods for Selecting a Random Sample
- Lesson 17：Sampling Variability

- Lesson 18：Sampling Variability and the Effect of Sample Size
- Lesson 19：Understanding Variability When Estimating a Population Proportion
- Lesson 20：Estimating a Population Proportion
- 海报要点：总体和样本；随机抽样；样本变异性；估计总体比例

D主题 比较总体（Topic D: Comparing Populations）

- 课程标准：7.SP.B.3, 7.SP.B.4
- Lesson 21：Why Worry About Sampling Variability?
- Lesson 22–23：Using Sample Data to Compare the Means of Two or More Populations
- 海报要点：用样本平均数比较两个总体

第6模块 几何（Module 6: Geometry）

- 来源：Teacher Edition PDF

A主题 未知角（Topic A: Unknown Angles）

- 课程标准：7.G.B.5
- Lesson 1：Complementary and Supplementary Angles
- Lesson 2–4：Solving for Unknown Angles Using Equations
- 海报要点：余角和补角；对顶角；列方程求角

B主题 作三角形（Topic B: Constructing Triangles）

- 课程标准：7.G.A.2
- Lesson 5：Identical Triangles
- Lesson 6：Drawing Geometric Shapes
- Lesson 7：Drawing Parallelograms
- Lesson 8：Drawing Triangles
- Lesson 9：Conditions for a Unique Triangle—Three Sides and Two Sides and the Included Angle
- Lesson 10：Conditions for a Unique Triangle—Two Angles and a Given Side
- Lesson 11：Conditions on Measurements That Determine a Triangle
- Lesson 12：Unique Triangles—Two Sides and a Non-Included Angle
- Lesson 13–14：Checking for Identical Triangles
- Lesson 15：Using Unique Triangles to Solve Real-World and Mathematical Problems
- 海报要点：画三角形和平行四边形；确定唯一三角形的条件

C主题 截立体图形（Topic C: Slicing Solids）

- 课程标准：7.G.A.3
- Lesson 16：Slicing a Right Rectangular Prism with a Plane
- Lesson 17：Slicing a Right Rectangular Pyramid with a Plane
- Lesson 18：Slicing on an Angle
- Lesson 19：Understanding Three-Dimensional Figures
- 海报要点：用平面截长方体、棱锥；截面形状
- 本地化/安全：截面演示用橡皮泥或泡沫块，切割由老师操作或用塑料刀

D主题 面积和表面积问题（Topic D: Problems Involving Area and Surface Area）

- 课程标准：7.G.B.6

- Lesson 20：Real-World Area Problems
- Lesson 21：Mathematical Area Problems
- Lesson 22：Area Problems with Circular Regions
- Lesson 23–24：Surface Area
- 海报要点：实际和数学面积问题；含圆的图形；表面积

E主题 体积问题（Topic E: Problems Involving Volume）

- 课程标准：7.G.B.6
- Lesson 25：Volume of Right Prisms
- Lesson 26：Volume of Composite Three-Dimensional Objects
- Lesson 27：Real-World Volume Problems
- 海报要点：直棱柱体积；组合体体积；实际问题

八年级（Grade 8）· 7 个模块 · 21 个 Topic

第1模块 整数指数和科学记数法（Module 1: Integer Exponents and Scientific Notation）

- 来源：Teacher Edition PDF

A主题 指数记法和整数指数的性质（Topic A: Exponential Notation and Properties of Integer Exponents）

- 课程标准：8.EE.A.1
- Lesson 1：Exponential Notation
- Lesson 2：Multiplication of Numbers in Exponential Form
- Lesson 3：Numbers in Exponential Form Raised to a Power
- Lesson 4：Numbers Raised to the Zeroth Power
- Lesson 5：Negative Exponents and the Laws of Exponents
- Lesson 6：Proofs of Laws of Exponents
- 海报要点：指数记法；同底数幂相乘；幂的乘方；零指数和负指数

B主题 数量级和科学记数法（Topic B: Magnitude and Scientific Notation）

- 课程标准：8.EE.A.3, 8.EE.A.4
- Lesson 7：Magnitude
- Lesson 8：Estimating Quantities
- Lesson 9：Scientific Notation
- Lesson 10：Operations with Numbers in Scientific Notation
- Lesson 11：Efficacy of Scientific Notation
- Lesson 12：Choice of Unit
- Lesson 13：Comparison of Numbers Written in Scientific Notation and Interpreting Scientific Notation Using Technology
- 海报要点：数量级；估计；科学记数法的运算；选择合适单位

第2模块 全等（Module 2: The Concept of Congruence）

- 来源：Teacher Edition PDF

A主题 基本刚体运动（Topic A: Definitions and Properties of the Basic Rigid Motions）

- 课程标准：8.G.A.1
- Lesson 1：Why Move Things Around?
- Lesson 2：Definition of Translation and Three Basic Properties
- Lesson 3：Translating Lines
- Lesson 4：Definition of Reflection and Basic Properties
- Lesson 5：Definition of Rotation and Basic Properties
- Lesson 6：Rotations of 180 Degrees
- 海报要点：平移、反射、旋转的定义和基本性质；旋转 180°

B主题 刚体运动的序列（Topic B: Sequencing the Basic Rigid Motions）

- 课程标准：8.G.A.2

- Lesson 7：Sequencing Translations
- Lesson 8：Sequencing Reflections and Translations
- Lesson 9：Sequencing Rotations
- Lesson 10：Sequences of Rigid Motions
- 海报要点：按顺序做平移、反射、旋转

C主题 全等与角的关系（Topic C: Congruence and Angle Relationships）

- 课程标准：8.G.A.2, 8.G.A.5
- Lesson 11：Definition of Congruence and Some Basic Properties
- Lesson 12：Angles Associated with Parallel Lines
- Lesson 13：Angle Sum of a Triangle
- Lesson 14：More on the Angles of a Triangle
- 海报要点：全等的定义；平行线截出的角；三角形内角和与外角

第3模块 相似（Module 3: Similarity）

- 来源：Teacher Edition PDF

A主题 位似（Topic A: Dilation）

- 课程标准：8.G.A.3
- Lesson 1：What Lies Behind “Same Shape”？
- Lesson 2：Properties of Dilations
- Lesson 3：Examples of Dilations
- Lesson 4：Fundamental Theorem of Similarity (FTS)
- Lesson 5：First Consequences of FTS
- Lesson 6：Dilations on the Coordinate Plane
- Lesson 7：Informal Proofs of Properties of Dilations (optional)
- 海报要点：位似中心和比例因子；相似基本定理；坐标平面上的位似

B主题 相似图形（Topic B: Similar Figures）

- 课程标准：8.G.A.4, 8.G.A.5
- Lesson 8：Similarity
- Lesson 9：Basic Properties of Similarity
- Lesson 10：Informal Proof of AA Criterion for Similarity
- Lesson 11：More About Similar Triangles
- Lesson 12：Modeling Using Similarity
- 海报要点：相似的定义和性质；AA 判定；用相似测量

第4模块 线性方程（Module 4: Linear Equations）

- 来源：Teacher Edition PDF

A主题 列方程和解线性方程（Topic A: Writing and Solving Linear Equations）

- 课程标准：8.EE.C.7
- Lesson 1：Writing Equations Using Symbols
- Lesson 2：Linear and Nonlinear Expressions in x
- Lesson 3：Linear Equations in x

- Lesson 4：Solving a Linear Equation
- Lesson 5：Writing and Solving Linear Equations
- Lesson 6：Solutions of a Linear Equation
- Lesson 7：Classification of Solutions
- Lesson 8：Linear Equations in Disguise
- Lesson 9：An Application of Linear Equations
- 海报要点：用符号列方程；线性方程；解的个数：一个、无解、无数个

B主题 二元线性方程及其图像（Topic B: Linear Equations in Two Variables and Their Graphs）

- 课程标准：8.EE.B.5
- Lesson 10：A Critical Look at Proportional Relationships
- Lesson 11：Constant Rate
- Lesson 12：Linear Equations in Two Variables
- Lesson 13：The Graph of a Linear Equation in Two Variables
- Lesson 14：The Graph of a Linear Equation—Horizontal and Vertical Lines
- 海报要点：恒定速率；二元一次方程；图像；水平线和竖直线

C主题 斜率和直线方程（Topic C: Slope and Equations of Lines）

- 课程标准：8.EE.B.5, 8.EE.B.6
- Lesson 15：The Slope of a Non-Vertical Line
- Lesson 16：The Computation of the Slope of a Non-Vertical Line
- Lesson 17：The Line Joining Two Distinct Points of the Graph y = mx + b Has Slope m
- Lesson 18：There Is Only One Line Passing Through a Given Point with a Given Slope
- Lesson 19：The Graph of a Linear Equation in Two Variables Is a Line
- Lesson 20：Every Line Is a Graph of a Linear Equation
- Lesson 21：Some Facts About Graphs of Linear Equations in Two Variables
- Lesson 22：Constant Rates Revisited
- Lesson 23：The Defining Equation of a Line
- 海报要点：斜率；两点求斜率；y = mx + b；直线的方程

D主题 线性方程组（Topic D: Systems of Linear Equations and Their Solutions）

- 课程标准：8.EE.B.5, 8.EE.C.8
- Lesson 24：Introduction to Simultaneous Equations
- Lesson 25：Geometric Interpretation of the Solutions of a Linear System
- Lesson 26：Characterization of Parallel Lines
- Lesson 27：Nature of Solutions of a System of Linear Equations
- Lesson 28：Another Computational Method of Solving a Linear System
- Lesson 29：Word Problems
- Lesson 30：Conversion Between Celsius and Fahrenheit
- Lesson 31：System of Equations Leading to Pythagorean Triples
- 海报要点：联立方程；图像的交点；平行线；消元法；应用题
- 本地化/安全：摄氏与华氏换算保留，作为公式应用的例子（中国日常用摄氏度）

第5模块 几何中的函数（Module 5: Examples of Functions from Geometry）

- 来源：Teacher Edition PDF

A主题 函数（Topic A: Functions）

- 课程标准：8.F.A.1, 8.F.A.2, 8.F.A.3
- Lesson 1：The Concept of a Function
- Lesson 2：Formal Definition of a Function
- Lesson 3：Linear Functions and Proportionality
- Lesson 4：More Examples of Functions
- Lesson 5：Graphs of Functions and Equations
- Lesson 6：Graphs of Linear Functions and Rate of Change
- Lesson 7：Comparing Linear Functions and Graphs
- Lesson 8：Graphs of Simple Nonlinear Functions
- 海报要点：函数的概念和定义；线性函数；函数图像；比较函数；非线性函数

B主题 体积（Topic B: Volume）

- 课程标准：8.G.C.9
- Lesson 9：Examples of Functions from Geometry
- Lesson 10：Volumes of Familiar Solids—Cones and Cylinders
- Lesson 11：Volume of a Sphere
- 海报要点：几何中的函数；圆锥、圆柱、球的体积
- 本地化/安全：装水实验擦干洒出的水，防止滑倒

第6模块 线性函数（Module 6: Linear Functions）

- 来源：Teacher Edition PDF

A主题 线性函数（Topic A: Linear Functions）

- 课程标准：8.F.B.4, 8.F.B.5
- Lesson 1：Modeling Linear Relationships
- Lesson 2：Interpreting Rate of Change and Initial Value
- Lesson 3：Representations of a Line
- Lesson 4–5：Increasing and Decreasing Functions
- 海报要点：建立线性模型；变化率和初始值；增函数和减函数

B主题 双变量数值数据（Topic B: Bivariate Numerical Data）

- 课程标准：8.SP.A.1, 8.SP.A.2
- Lesson 6：Scatter Plots
- Lesson 7：Patterns in Scatter Plots
- Lesson 8：Informally Fitting a Line
- Lesson 9：Determining the Equation of a Line Fit to Data
- 海报要点：散点图；散点图中的规律；拟合直线及其方程

C主题 线性和非线性模型（Topic C: Linear and Nonlinear Models）

- 课程标准：8.SP.A.1, 8.SP.A.2, 8.SP.A.3
- Lesson 10：Linear Models

- Lesson 11：Using Linear Models in a Data Context
- Lesson 12：Nonlinear Models in a Data Context (Optional)
- 海报要点：线性模型；在数据情境中用线性模型；非线性模型

D主题 双变量分类数据（Topic D: Bivariate Categorical Data）

- 课程标准：8.SP.A.4
- Lesson 13：Summarizing Bivariate Categorical Data in a Two-Way Table
- Lesson 14：Association Between Categorical Variables
- 海报要点：双向表；分类变量之间的关联

第7模块 用几何认识无理数（Module 7: Introduction to Irrational Numbers Using Geometry）

- 来源：Teacher Edition PDF

A主题 平方根和立方根（Topic A: Square and Cube Roots）

- 课程标准：8.NS.A.1, 8.NS.A.2, 8.EE.A.2
- Lesson 1：The Pythagorean Theorem
- Lesson 2：Square Roots
- Lesson 3：Existence and Uniqueness of Square Roots and Cube Roots
- Lesson 4：Simplifying Square Roots (Optional)
- Lesson 5：Solving Equations with Radicals
- 海报要点：勾股定理；平方根；平方根和立方根的存在与唯一；解含根号的方程

B主题 数的小数展开（Topic B: Decimal Expansions of Numbers）

- 课程标准：8.NS.A.1, 8.NS.A.2, 8.EE.A.2
- Lesson 6：Finite and Infinite Decimals
- Lesson 7：Infinite Decimals
- Lesson 8：The Long Division Algorithm
- Lesson 9：Decimal Expansions of Fractions, Part 1
- Lesson 10：Converting Repeating Decimals to Fractions
- Lesson 11：The Decimal Expansion of Some Irrational Numbers
- Lesson 12：Decimal Expansions of Fractions, Part 2
- Lesson 13：Comparing Irrational Numbers
- Lesson 14：Decimal Expansion of π
- 海报要点：有限和无限小数；长除法；循环小数化分数；无理数的小数；π

C主题 勾股定理（Topic C: The Pythagorean Theorem）

- 课程标准：8.G.B.6, 8.G.B.7, 8.G.B.8
- Lesson 15：Pythagorean Theorem, Revisited
- Lesson 16：Converse of the Pythagorean Theorem
- Lesson 17：Distance on the Coordinate Plane
- Lesson 18：Applications of the Pythagorean Theorem
- 海报要点：勾股定理再探；逆定理；坐标平面上的距离；应用
- 本地化/安全：英尺、英寸改为米、厘米

D主题 根式的应用（Topic D: Applications of Radicals and Roots）

- 课程标准：8.G.B.7, 8.G.C.9
- Lesson 19：Cones and Spheres
- Lesson 20：Truncated Cones
- Lesson 21：Volume of Composite Solids
- Lesson 22：Average Rate of Change
- Lesson 23：Nonlinear Motion
- 海报要点：圆锥和球；圆台；组合体体积；平均变化率；非线性运动